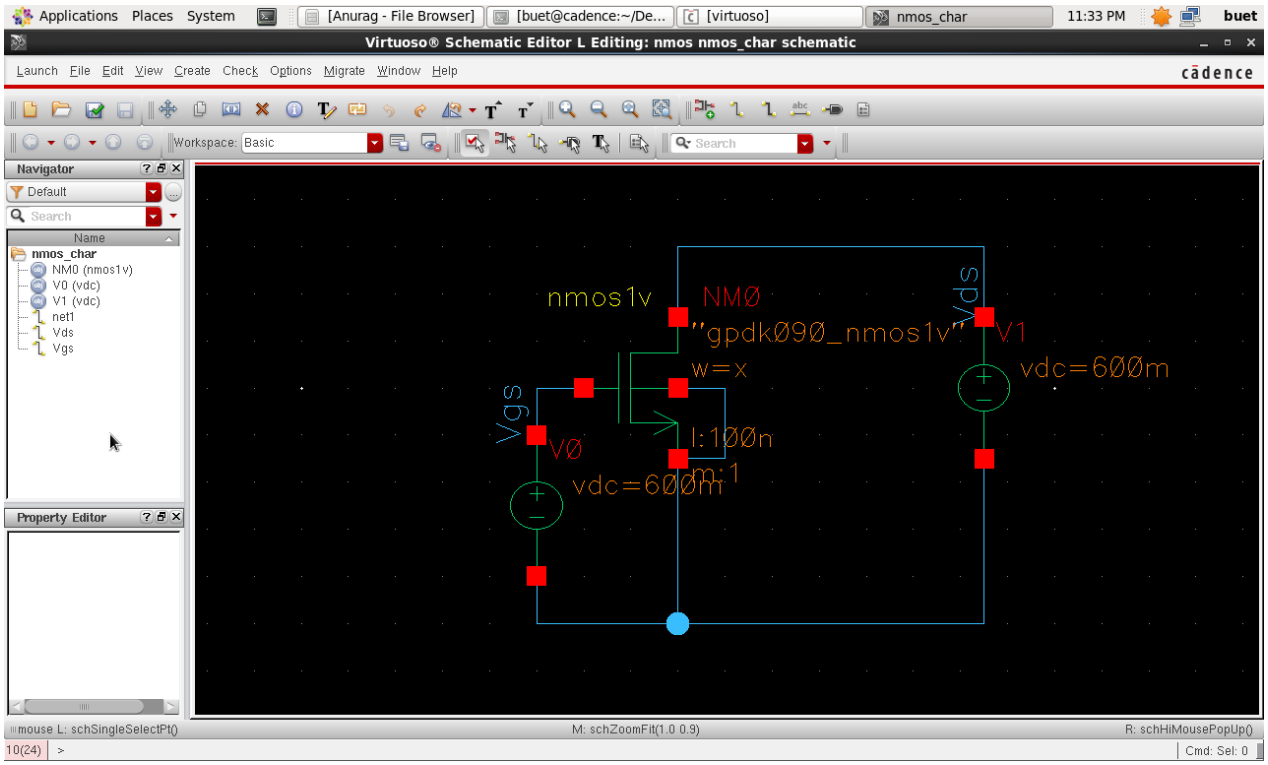
Assignment - 01

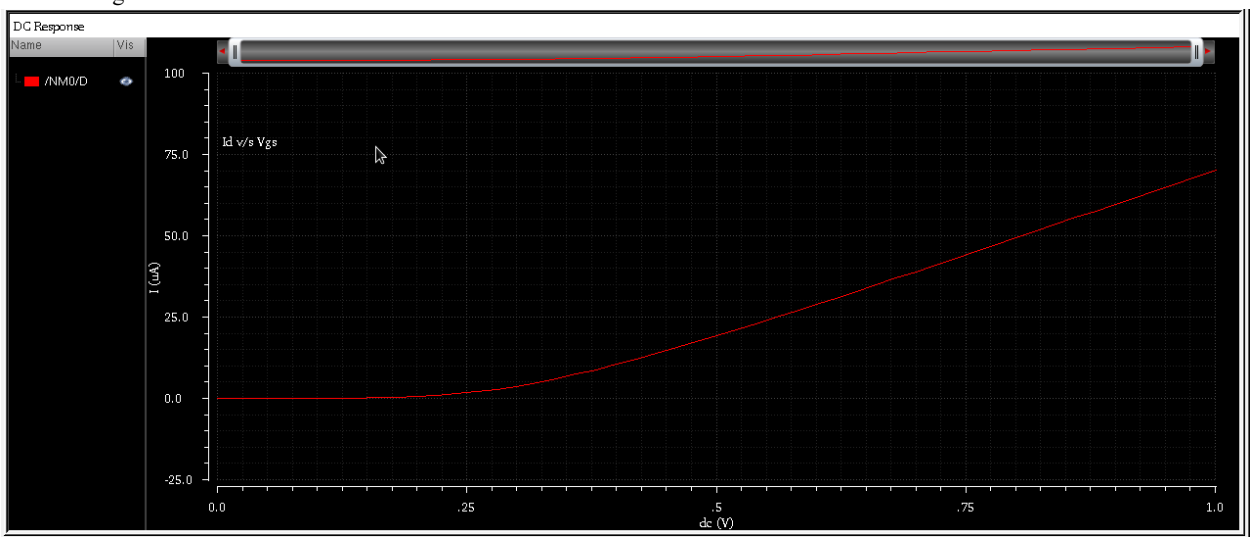
DC analysis of NMOS

1. Schematic diagram of NMOS:



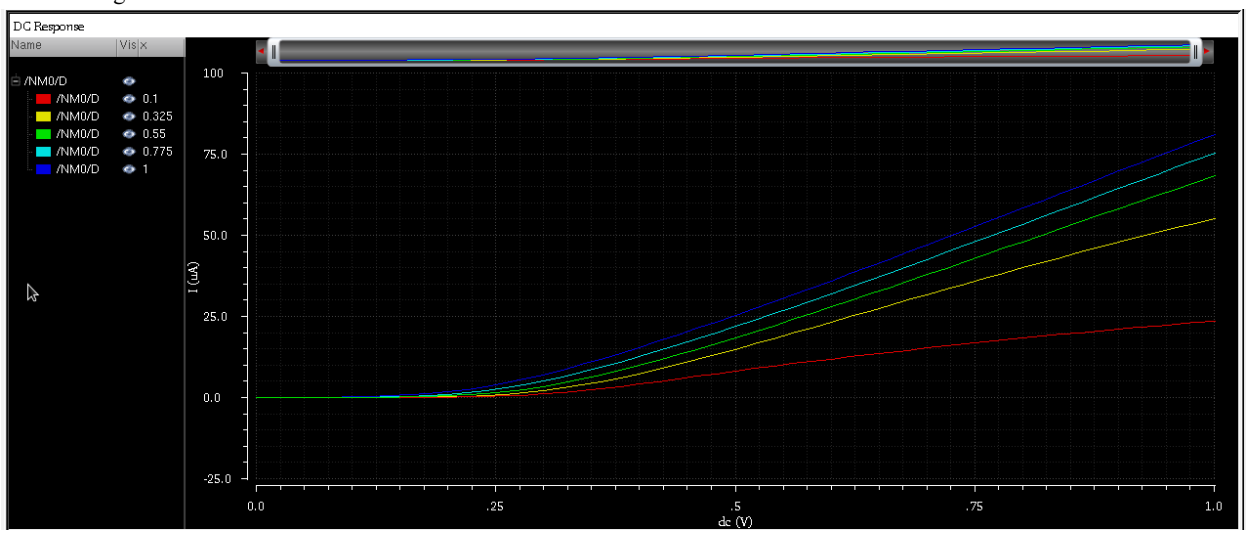
2. I-V Characteristics:

- I_d v/s V_{gs}:



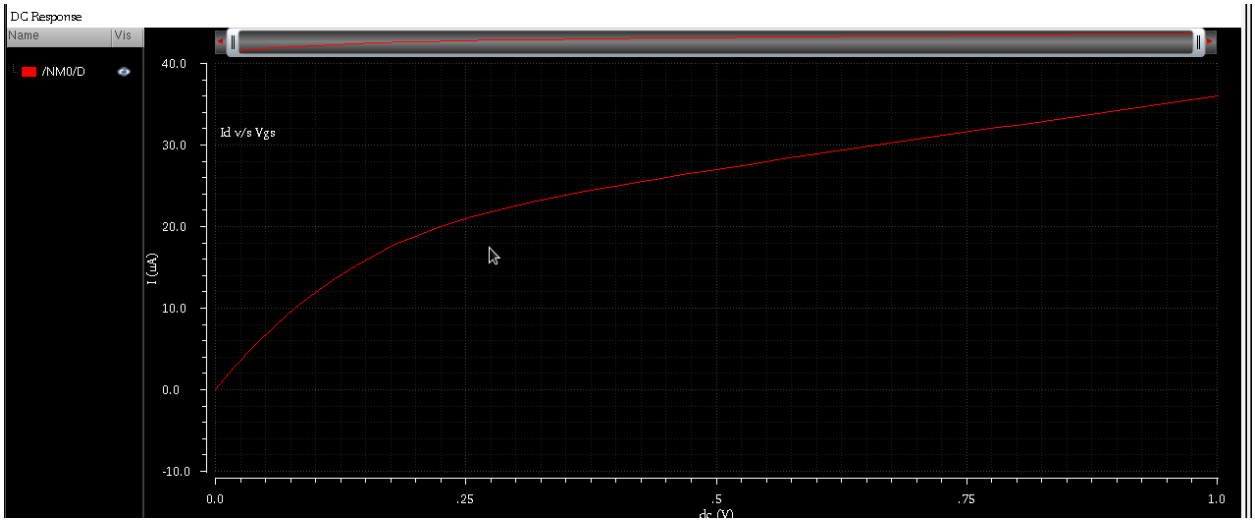
- The drain current (I_d) remains nearly zero until the gate voltage (V_{gs}) reaches the threshold voltage (V_{th})
- Beyond V_{th}, I_d increases rapidly with V_{gs} as a strong inversion channel is formed in the NMOS transistor.

- I_d v/s V_{gs} w. r. t. different of V_{ds}:



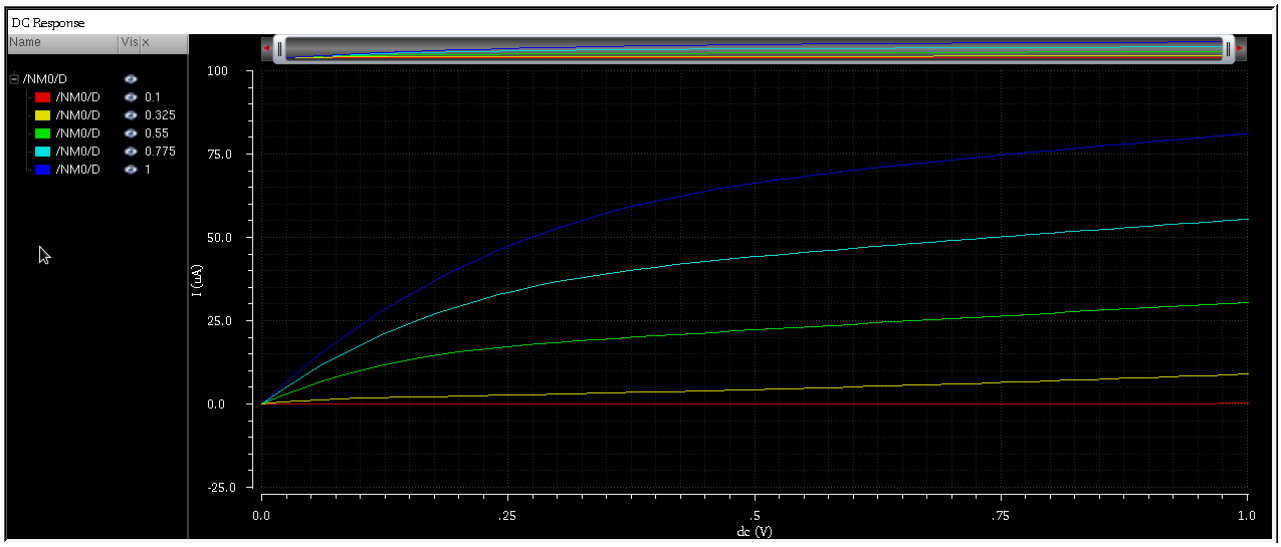
  - For a fixed V_{gd}, the drain current increases as V_{ds} increases because a higher drain voltage creates a stronger electric field, allowing more carriers to flow through the channel until saturation is approached.

- I_d v/s V_{ds}:



  - The drain current (I_d) increases with increasing drain-to-source voltage (V_{ds}) while the gate voltage is kept constant."
  - At higher V_{ds}, the rate of increase of I_d becomes smaller as the NMOS transistor approaches the saturation region.

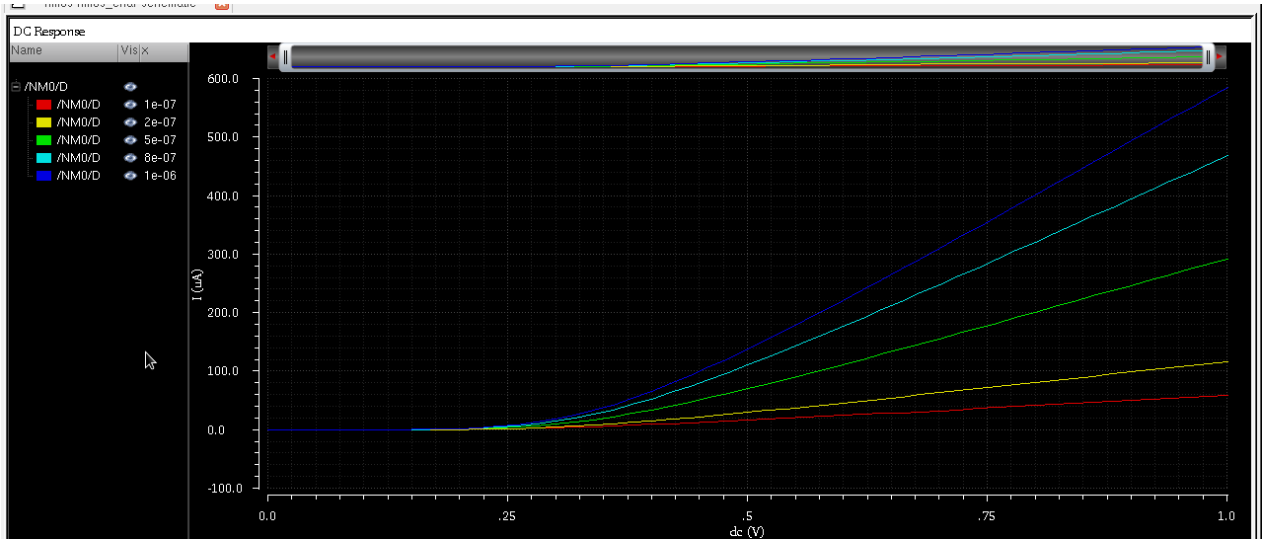
- I_d v/s V_{ds} w. r. t. different of V_{gs}:



- The drain current (ID) increases with increasing drain-to-source voltage (VDS) and enters the saturation region after the pinch-off condition is reached.
- At higher VDS, the drain current increases only slightly due to channel length modulation, indicating the practical behavior of the NMOS transistor.

3. I-V Characteristics w. r. t. Variation of Parameter:

- I_d-V_{gs} Characteristics w. r. t. Variation of Width:



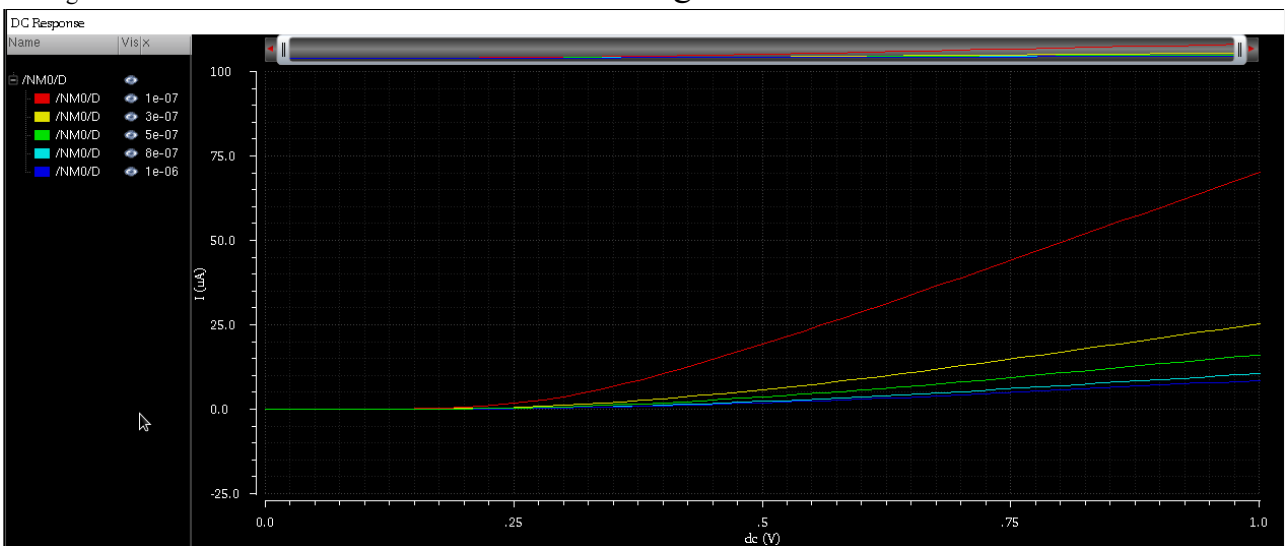
- As the channel width (W) increases, the drain current increases significantly because a wider channel provides a larger conduction path for charge carriers.
- The reason comes from the MOSFET current equation.

  For an NMOS in saturation,

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$$

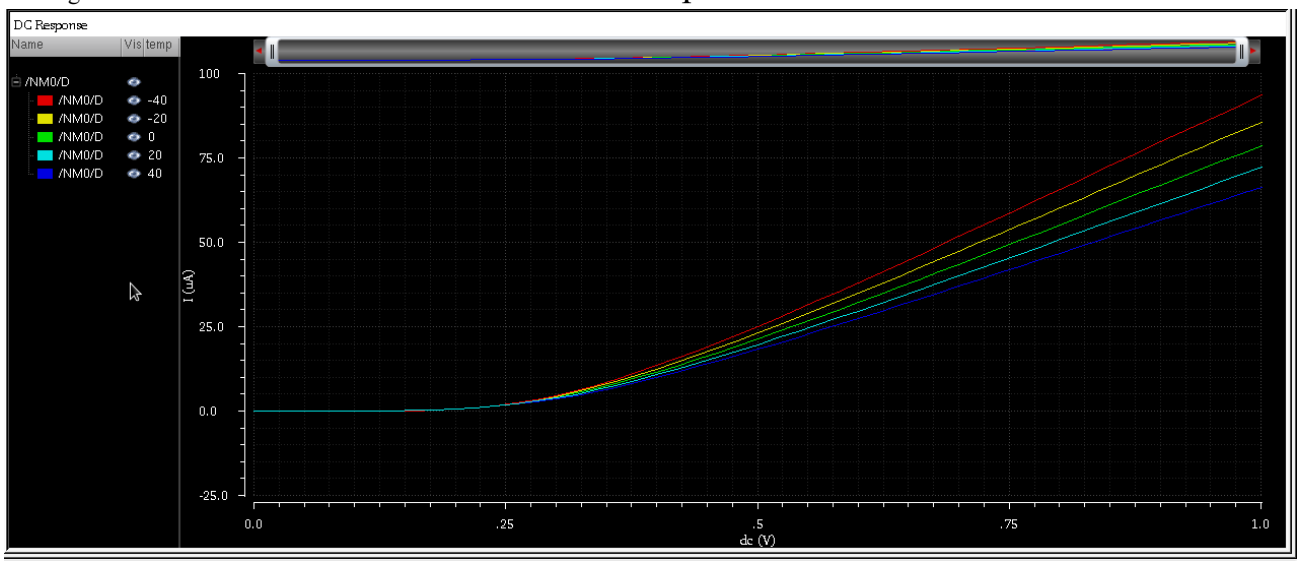
  Here, $I_D \propto \frac{W}{L}$.

- I_d-V_{gs} Characteristics w. r. t. Variation of Length:



  - As the channel length (l) increases, the drain current decreases because the channel resistance increases, reducing the flow of charge carriers.

- I_d-V_{gs} Characteristics w. r. t. Variation of Temperature:



  - As the temperature increases, the drain current decreases because carrier mobility reduces due to increased lattice scattering, which lowers the channel conductivity.